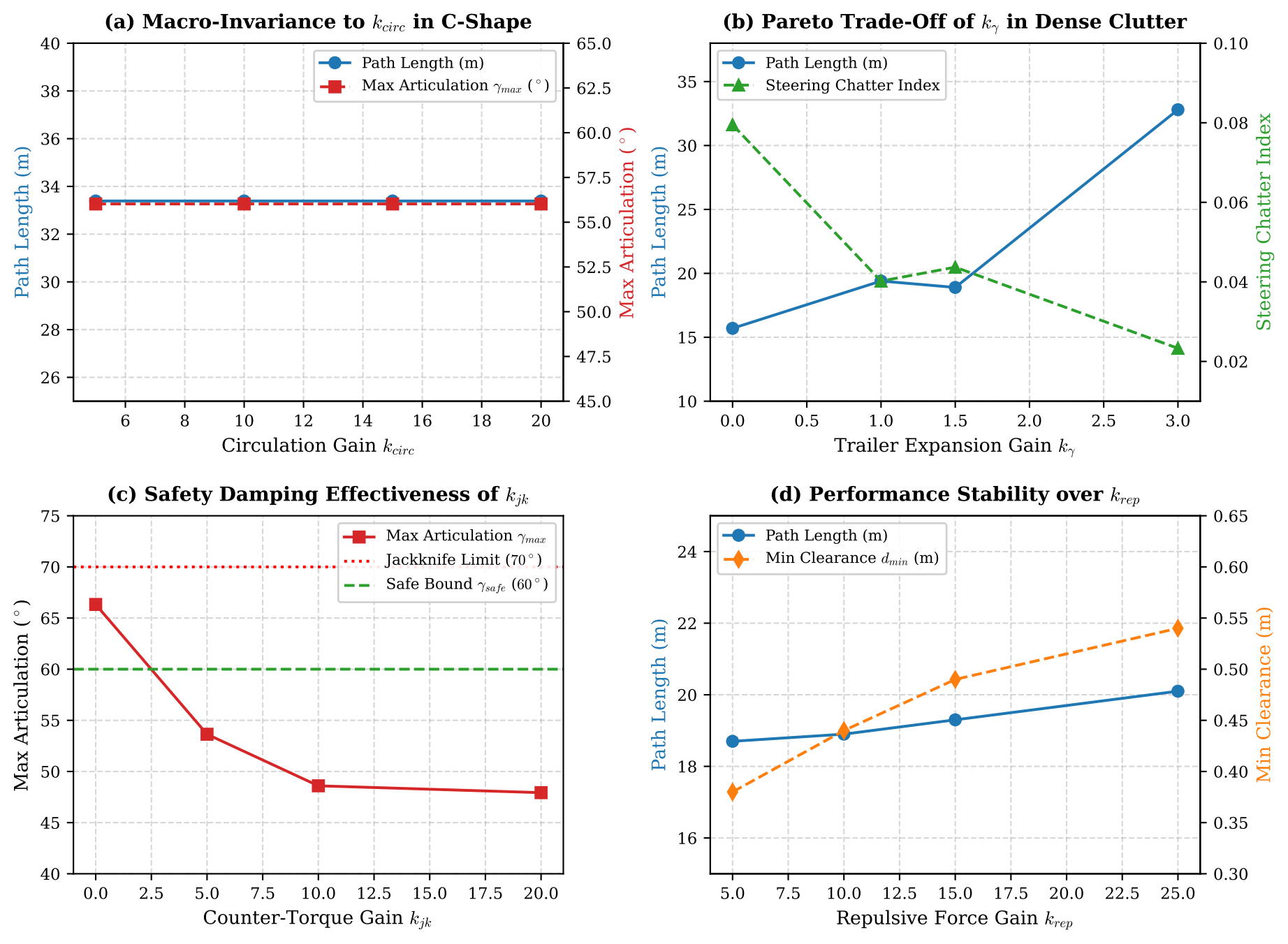
(a) Macro-Invariance to k_{circ} in C-Shape
Path Length (m)
Max Articulation γ_{max} (°)
Path Length (m)
Max Articulation (°)
Circulation Gain k_{circ}
(b) Pareto Trade-Off of k_γ in Dense Clutter
Path Length (m)
Steering Chatter Index
Path Length (m)
Steering Chatter Index
Trailer Expansion Gain k_γ
(c) Safety Damping Effectiveness of k_{jk}
Max Articulation γ_{max}
Jackknife Limit (70°)
Safe Bound γ_{safe} (60°)
Max Articulation (°)
Counter-Torque Gain k_{jk}
(d) Performance Stability over k_{rep}
Path Length (m)
Min Clearance d_{min} (m)
Path Length (m)
Min Clearance (m)
Repulsive Force Gain k_{rep}